MODERATED LEARNING MANAGEMENT SYSTEM

PROJECT MANAGEMENT SYSTEM PLAN

PHASE 1: PROJECT PLANNING AND MANAGEMANT

CS4433 SOFTWARE ENGINEERING

Prepared by:

Khauhelo Mosanteli

202322649 Bsc. Computer Science

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1. Abstract

This document presents the Project Management Plan for the development of a Moderated Learning Management system (LMS) for a secondary school. The purpose of the system is to support the management of Users, courses, assignments, submissions, marking and moderation.

The first phase focuses on determining whether the proposed system is feasible and planning how the project will be completed within the available seven-week period. The feasibility study considers technical, operational, economic, legal and schedule feasibility. A work breakdown structure and project schedule are also provided to divided the project into manageable activities and identity dependencies, milestones and critical path.

The document also identifies potential technical, project and people-related risks. Each risk is assessed according to its like likelihood, impact and exposure, with a mitigation strategy and responsible owner assigned. The risk register will be reviewed and updated at each phase gate.

Based on the feasibility assessment, the proposed LMS is considered to develop, provided that the project is carefully managed, security and data protection are considered, and the identified risks are monitored throughout the project.

2. Introduction

This project involves the development of a Moderated Learning Management System for a secondary school. The purpose of the system is to help the school manage users, courses, assignments, submissions and assessments marks.

The system will have four main type of users: Students, Lecturers, Heads of Department and a System Administrator. Each user will have different permissions depending on their role. For instance, the lecturers will create assignments, mark submissions and provide feedback, while HoDs will review and moderate marks before they are released to students.

The system will be developed as a C++17 application with SQLite used for persistent data storage. It will run on a school computer or local network and will not depend on an internet connection, because the school has unreliable internet access according to the case study.

This project will follow the waterfall development model with V-model verification and validation. The project has a duration of seven weeks, so proper planning is important to make sure the requirements, design, implementation, testing and final maintenance activities are completed within the available time.

The purpose of this Project management Plan is to determine whether the project is feasible, break the project in manageable activities, create a schedule, identify important dependencies and milestones, and identify risks that could affect the successful completion of the project.

3. Feasibility Study

The feasibility study considers whether the proposed LMS can realistically be developed and used. Five areas have been considered: technical, operational, economic, legal and schedule feasibility.

3.1 Technical Feasibility

The application will be developed using c++17, while SQLite will be used to store information such as users, courses, assignments, submissions and marks. Cmake will be used to build the application and Git will be used for version control. A testing framework such as GoogleTest ot Catch2 will be used for automated testing, while static analysis tool such as cppcheck orc clang-tidy can be used to identify possible problems in the code.

The system is not required to have a web interface or cloud server. It will run on a school computer or local network. This makes the technical requirements more manageable and also supports the requirement that the system should work when the internet connection is unreliable.

3.2 Operational Feasibility

The proposed system is operationally feasible because its functions are based on activites that normally take place in a secondary school

The system should support 4 different user roles. The administrator will manage users, departments, courses, lectures and student enrolments. Lecturers will create assignments, manage deadlines, provide extensions, mark submissions and provide feedback. HoDs will review and moderate marks before they are released. Students will view their courses and assignments, submit their work and view released marks and feedback.

One other issue is that the principal stated that teachers are not IT experts, therefore the system should have a simple understandable interface and should avoid unnecessary complexities.

Another operational requirement is that the system must work offline because the school's internet connection is unreliable. This means the normal school activities should not depend on internet connection.

Based on these considerations, the LMS is operationally feasible because it supports the school's activities and takes the user's needs and technical environment into account.

3.3 Economic feasibility

The proposed system is economically feasible based on the information given in the case study

The system does not require expensive online services or a cloud platform because it will operate on a school computer or local network. SQLite can provide the required database functionality without requiring a separate database server.

The development tools required for the project are suitable for development process and do not require the project to depend on expensive infrastructure.

Based on the available information, there is no identified factors that would prevent the development of the proposed LMS.

3.4 Legal feasibility

The LMS will store personal and academic information about students and staff. Therefore, privacy, data protection and security must be considered during the development and use of the system.

Assessment information must also be protected. The system will maintain an audit record of changes to marks, including who made the change, when it was made, the previous mark and the new mark and reason for change.

Passwords must not be stored as plain text. The system will use proper password hashes using an appropriate recognized library.

Subject to these requirements being addressed, the project is considered legally feasible.

3.5 Schedule Feasibility

The project has a duration of seven weeks.

The planned phases are:

- Week 1: project planning and management
- Week 1-2: requirements engineering
- Week 2-3: system design
- Week 4-5: implementation
- Week 5-6: verification and validation
- Week 6: deployment
- Week 7: maintenance and project completion

3.6 Feasibility recommendation

Based on the technical, economic, legal, operational and schedule assessments, the proposed

LMS is considered feasible to develop.

4. Work breakdown structure

1. Project planning and Management
    - Feasibility study
    - Work breakdown structure
    - Project schedule
    - Risk management

2. Requirements engineering
    - Stakeholder identification
    - Stakeholder analysis
    - Conflict and ambiguity resolution
    - Software requirements specification
    - Functional requirements
    - Non- functional requirements
    - Use case diagrams
    - Fully dressed use diagrams

3. System Design
    - Architecture selection
    - Component diagram
    - Class diagram
    - Design patterns
    - State machine diagram
    - State machine diagram

- Activity diagram
- Sequence diagrams
- Database design
- Entity relationship diagram
- Data dictionary
- SQL schema

4. implementation

- Project setup
- User and authentication system
- Role based access
- Course management
- Submission management
- Assignment management
- Extension and resubmission functionality
- Marking and feedback
- Moderation and mark release
- SQLite data access
- Security and input validation

5. Verification and validation

- Test strategy
- Unit testing
- Integration testing
- System testing

- Acceptance testing
- Defect logging
- Test coverage

6. Configuration Management and Quality assurance

- Git branching strategy
- Continuous integration
- Static analysis
- Code review
- Code quality metrics

7. deployment

- Deployment diagram
- Installation guide
- User manuals
- Release notes
- Version 1.0 release

8. Maintenance and completion

- Analyse change request CR1
- Impact analysis
- Implement version 1.1
- Regression testing
- Update documentation
- Final report
- Professional, legal and ethical issues

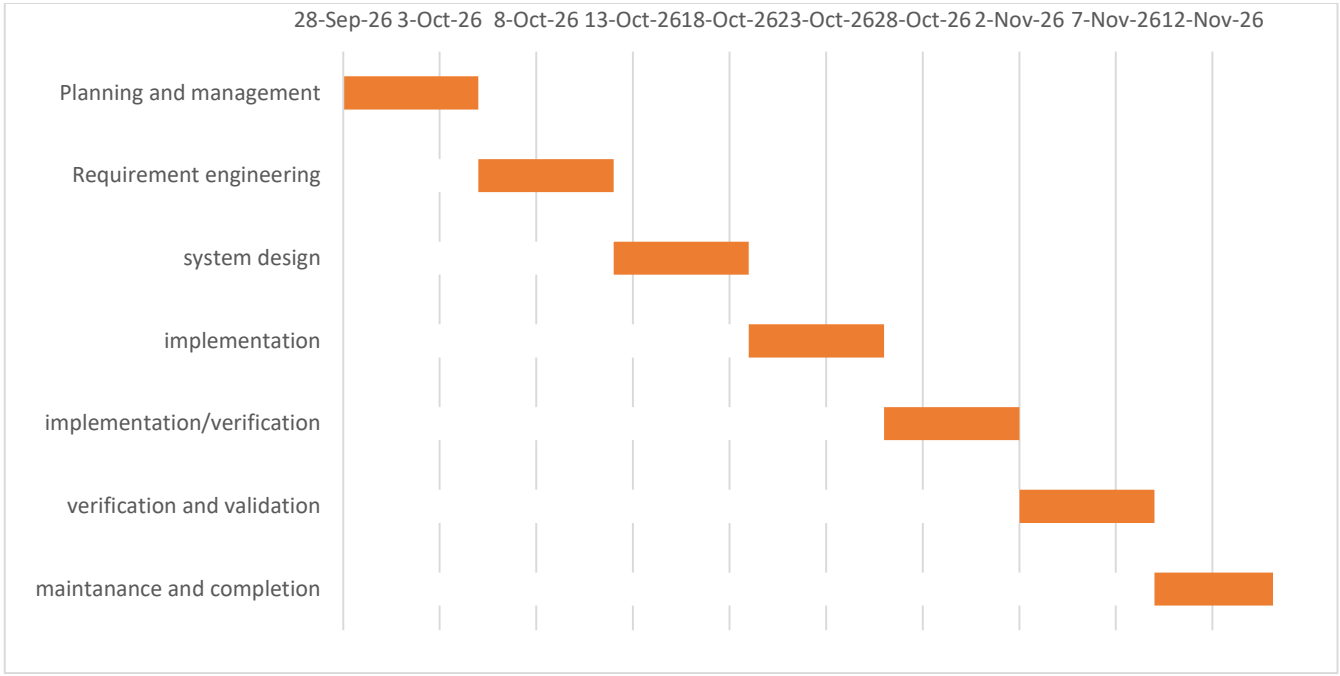
- Viva preparation

5. Project schedule

The project will be completed in 7 weeks.

The project brief specifies seven weeks, with requirements spanning week 1-2, design weeks 2-3, implementation weeks 4-5, verification weeks 5-6, deployment week 6, and maintenance/completion week 7.

Phase 1: Project planning and Management



6. Dependencies

Main dependency flow

Feasibility -> system design -> gate 2 ->implementation -> testing ->

Deployment -> maintenance -> completion

The requirements must be sufficiently defined before the detailed system design can be completed.

The system design must then pass Gate 2

Before implementation can begin.

Implementation must be completed to sufficient level before full system testing can take place. Testing must be completed before the system is released as version 1.0.

The maintenance change request is performed after the initial release and results in Version 1.1

The major dependency that could have the greatest effect on the schedule is the relationship between system desing and implementation because implementation cannot begin until the design has passed Gate 2.

7. project Milestones

The following milestones will be used to monitor the progress of the project

Mile	Target week	Expected Result
M1 – project planning completed	Week 1	Project Management Plan Completed
Requirements completed	Week 2	SRS and use model completed
Gate 1	Week 2	Requirements reviewed
System design	Week3	Design document completed
Gate 2	Week 3	Design approved before implementation
Implementation completed	Week 5	Main LMS functionality implementation
Gate 3	Week 6	Testing and validation reviewed
Version 1.0	Week 6	Initial release prepared
Version 1.1	Week 7	Change request implemented
Project completion	Week 7	Final report and viva preparation completed

8. Critical Path

The critical path is the sequence of activities that must be completed on to avoid delayed the overall project.

For this project, the main critical path is:

Requirements Engineering -> system design -> Gate 2-> implementation -> testing -> deployment -> maintenance -> final completion

Requirements engineering is important because the design depends on having sufficiently clear requirements. The implementation phase depends on the design being completed and approved at Gate 2. Testing depends on having a working implementation, while deployment depends on the system being sufficiently tested.

Because of these dependencies, delays in requirements engineering, system design, implementation or testing could have a direct effect on the final completion of the project.

The team will therefore monitor these activities closely and review their progress at each phase gate.

9. Risk Management

Risk management will be used throughout the project to identify possible problems before they have a major effect on the project.

The risk register will be reviewed and updated at each phase gate. New risks will be added when necessary, while the likelihood and impact of existing risks will be reassessed as the project progresses.

For this project, likelihood and impact are rated from 1 to 5.

Exposure = Likelihood × Impact

A higher exposure indicates that the risk requires greater attention.

10. Risk Register

ID	RISK	TYPE	LIKEHOOD	IMPACT	EXPOSURE	MITIGATION	OWNER
R01	Requirements are misunderstood or incomplete	Project	3	5	15	Review stakeholder statements carefully and clarify conflicts before finalizing requirements	Requirements lead
R02	The project fails behind the seven week schedule	Project	4	5	20	Monitor weekly progress and address delayed activities as early as possible	Project Manager

R03	Problems occur when connecting the c++ application to SQLite	Technical	3	4	12	Test database operations early and perform integration testing throughout development	Database lead
R04	Student information or password are not adequately protected	Technical	2	5	10	Use salted password hashes, parameterized queries, input validation and role-based access	Security lead
R05	A team member does not complete assigned work on time	People	3	4	12	Assign clear responsibilities, monitor progress and redistribute work when necessary	Project Manager
R06	The implementation does not match the UML design	Technical	3	4	12	Review the implementation against the approved design and	Design lead

						maintain traceability	
R07	Testing is delayed until too late in project	Project	3	5	15	Begin automated testing during implementation rather than waiting until the end	Testing lead
R08	Project files or source code are lost	Technical	2	5	10	Use Git regularly and push work to the hosted repository	Configuration lead
R09	Users find the system difficult to use	People	3	3	9	Keep the system simple and use user testing and user manuals	Requirements lead
R10	The design does not pass Gate 2 on time	project	2	5	10	Review the design internally before Gate 2 and ensure it is consistent with the requirements	Design lead

Risk review plan

The risk register will not remain static. It will be reviewed:

- At the end of week 1
- At Gate 1
- At Gate 2
- At Gate 3
- Before final submission

New risks will be added when identified, while risks that are no longer relevant will be closed or updated.

11. Conclusion

This project management plan establishes the Foundation for developing the Moderated Learning Management System

The feasibility study indicates that the system is technically, operationally, economically, legally and schedule feasible, provided that the project id carefully managed. The work breakdown structure divides the project into manageable activities, while the Gantt chart provides a seven-week schedule.

The dependency analysis identifies the order in which major activities must be completed, with the requirements and design acting as important prerequisites for implementation. The critical path highlights activities where delays could affect the overall project completion date.

The risk register identifies the major technical, project and people-related risks and provides mitigation actions for each one. The register will be updated at the project’s phase gates.

Overall, the project can proceed to requirements engineering, with particular attention to requirements clarification, security, database design, testing and schedule control.

Phase 1: Project planning and Management